The Electroweak Scale from Recognition Science: Hierarchy Dissolution via the φ-Ladder

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

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Abstract

We derive the electroweak scale and dissolve the hierarchy problem within Recognition Science (RS). All mass scales in RS are positions on the φ-ladder ($\varphi = (1+\sqrt{5})/2$), with no bare parameters. The Higgs vacuum expectation value $v \approx 246$ GeV occupies rung $r_v \approx 27$ relative to the electron mass, while the Planck scale occupies rung $r_P \approx 107$. The electroweak–Planck hierarchy $v/M_P \approx \varphi^{-80} \approx 10^{-17}$ is an 80-rung structural gap—a moderate integer determined by the geometry of the φ-ladder, not a fine-tuning. Radiative corrections do not destabilize the rung position because RS has no bare parameters: masses are structural outputs of the cost functional, not inputs subject to renormalization. No new particles (superpartners, technifermions, KK excitations) are required.

1 Introduction

The hierarchy problem [**?**, **?**] is the question: why is the electroweak scale $v \sim 10^2$ GeV so much smaller than the Planck scale $M_P \sim 10^{19}$ GeV? In the Standard Model (SM), the Higgs boson mass receives quadratically divergent radiative corrections:

$$\delta m_H^2 \sim \frac{\Lambda^2}{16\pi^2}\left(6y_t^2 - 3g^2 - g'^2 - 6\lambda\right), \tag{1}$$

where Λ is the UV cutoff. If $\Lambda \sim M_P$, the corrections are $\delta m_H^2 \sim 10^{38}$ GeV^2, requiring a cancellation of 1 part in 10^{34} against the bare Higgs mass-squared parameter to achieve $m_H = 125$ GeV [**?**].

Proposed solutions include:

- **Supersymmetry (SUSY)** [**?**]: Cancels quadratic divergences via superpartner loops. Requires ≥ 105 new parameters. No superpartners found at the LHC up to ~ 2 TeV.
- **Composite Higgs** [**?**]: The Higgs is a pseudo-Nambu–Goldstone boson of a new strong sector. Constrained by precision electroweak data.

- **Extra dimensions** [?]: The Planck scale is lowered by geometric dilution. Constrained by LHC searches.
- **Anthropic landscape** [?]: The hierarchy is a selection effect among $\sim 10^{500}$ string vacua. Unfalsifiable.

RS [?] offers a different resolution: the hierarchy is *dissolved*, not solved. There is no fine-tuning because there are no bare parameters.

2 Mass Scales on the φ-Ladder

Definition 2.1 (φ-Ladder Positions)**.** In RS, every mass scale is a position on the φ-ladder. For any mass m, its rung position relative to a reference mass m_0 is:

$$r = \frac{\ln(m/m_0)}{\ln \varphi}. \tag{2}$$

Proposition 2.2 (Electroweak and Planck Rungs)**.** *Taking the electron mass* $m_e = 0.511$ MeV *as reference:*

$$r_v = \log_\varphi(v/m_e) = \frac{\ln(246{,}220/0.511)}{\ln(1.618)} \approx 27.2, \tag{3}$$

$$r_P = \log_\varphi(M_P/m_e) = \frac{\ln(1.22 \times 10^{22}/0.511)}{\ln(1.618)} \approx 107.1. \tag{4}$$

The hierarchy gap:

$$\Delta r = r_P - r_v \approx 80 \text{ rungs}. \tag{5}$$

Theorem 2.3 (Hierarchy as Rung Gap)**.** *The electroweak–Planck hierarchy is:*

$$\boxed{\frac{v}{M_P} = \varphi^{-\Delta r} = \varphi^{-80} \approx 1.9 \times 10^{-17}.} \tag{6}$$

Remark 2.4. The number $\Delta r = 80$ is a moderate integer, comparable to other structural integers in the theory (e.g., $D = 3$, $E = 12$, $V = 8$, $W = 17$). There is nothing unnatural about an 80-rung gap; it is the structural distance between two positions on the ladder.

3 Dissolution of the Hierarchy Problem

Theorem 3.1 (No Bare Parameters)**.** *In RS, the Higgs VEV is not a bare parameter plus radiative corrections. It is a structural output of the J-cost minimization.*

Proof. In the SM, the Higgs potential is:

$$V(\phi) = -\mu^2|\phi|^2 + \lambda|\phi|^4, \tag{7}$$

where μ^2 is the bare mass-squared parameter that receives quadratic corrections from UV physics. The hierarchy problem is: why is $\mu^2_{\text{phys}} \ll \Lambda^2$?

In RS, the potential $V(\phi)$ *emerges* from the J-cost functional of the ledger. The VEV $v = \langle\phi\rangle$ is not an input parameter but the solution to the cost minimization:

$$\frac{\partial J_{\text{EW}}}{\partial v} = 0 \quad \Rightarrow \quad v \text{ at rung } r_v. \tag{8}$$

Since v is an output (the position where the J-cost has its minimum), there is no "bare value" to protect from radiative corrections. The concept of fine-tuning is inapplicable: one cannot fine-tune a structural output. □

Remark 3.2 (Analogy with Atomic Energy Levels). The hydrogen atom energy levels $E_n = -13.6\ \text{eV}/n^2$ are structural solutions of the Schrödinger equation. They are not destabilized by vacuum fluctuations (the Lamb shift provides a small calculable correction, not a destabilization). Similarly, the φ-rung positions are structural solutions of the cost minimization.

3.1 UV Finiteness of the Ledger

Proposition 3.3 (No Quadratic Divergences)**.** *The recognition ledger is fundamentally discrete (8-tick epochs, finite voxel structure). There is no UV continuum limit that generates quadratic divergences.*

Proof. Quadratic divergences arise from integrating loop momenta up to an arbitrarily high cutoff Λ:

$$\int_0^\Lambda \frac{d^4k}{(2\pi)^4} \frac{1}{k^2 - m^2} \sim \Lambda^2. \tag{9}$$

In RS, the ledger has a natural UV scale set by the tick duration τ_0 and the voxel spacing $\ell_0 = c\tau_0$. Momenta above $k_{\max} = \pi/\ell_0$ do not exist in the ledger. The integral is naturally regulated:

$$\int_0^{k_{\max}} \frac{d^4k}{(2\pi)^4} \frac{1}{k^2 - m^2} \sim k_{\max}^2 \sim \ell_0^{-2}, \tag{10}$$

which is finite and corresponds to the highest rung of the φ-ladder. There is no "hierarchy" between this natural cutoff and the electroweak scale because the cutoff is itself a rung position on the ladder. □

4 Electroweak Masses from the VEV

Proposition 4.1 (Gauge Boson Masses)**.** *Given the VEV $v \approx 246$ GeV, the electroweak gauge boson masses follow from standard electroweak symmetry breaking:*

$$m_W = \frac{gv}{2} \approx 80.4\ \text{GeV}, \tag{11}$$

$$m_Z = \frac{m_W}{\cos\theta_W} \approx 91.2\ \text{GeV}, \tag{12}$$

where g is the $SU(2)$ gauge coupling and θ_W is the Weinberg angle (both derived from the $D = 3$ cube geometry).

Proposition 4.2 (Higgs Mass)**.** *The Higgs mass is:*

$$m_H = v\sqrt{2\lambda} \approx 125\ \text{GeV}, \tag{13}$$

where λ is the quartic coupling determined by the J-cost curvature at the VEV minimum.

Remark 4.3. The measured Higgs mass $m_H = 125.25 \pm 0.17$ GeV [**?**] is consistent with the RS prediction. The quartic coupling $\lambda \approx 0.13$ is not a free parameter but is determined by the cost functional.

5 Why No New Particles at the TeV Scale

Theorem 5.1 (Desert Prediction)**.** *RS predicts no new particles between the electroweak scale and the Planck scale beyond those already in the Standard Model.*

Proof. In the SM with SUSY, new particles near the TeV scale cancel quadratic divergences. In RS, there are no quadratic divergences (Proposition **??**), so no cancellation mechanism is needed. The φ-ladder rungs between $r_v \approx 27$ and $r_P \approx 107$ are occupied by the known SM particles (quarks, leptons, gauge bosons) at their respective rungs—no new particle species fill these intermediate rungs. □

Remark 5.2. The LHC null results for SUSY ($m_{\tilde{g}} > 2.3$ TeV, $m_{\tilde{q}} > 1.8$ TeV) [**?**], extra dimensions, and composite Higgs particles are consistent with the RS prediction. Future colliders (FCC-hh at 100 TeV) should continue to find no new particles.

6 Comparison with Other Solutions

Approach	**New particles**	**Free params**	**LHC status**	**Λ role**
SUSY	≥ 12	≥ 105	Not found	Cancel
Composite Higgs	Yes	Several	Constrained	Composite
Extra dims	Yes (KK)	Several	Constrained	Dilute
Anthropic	No	10^{500} vacua	N/A	Select
RS φ-ladder	**No**	**0**	**Consistent**	**Absent**

Remark 6.1. The key distinction: SUSY, composite Higgs, and extra dimensions *solve* the hierarchy problem by introducing new physics that cancels or screens the quadratic divergence. RS *dissolves* it by removing the conceptual framework (bare parameters + radiative corrections) in which the problem is formulated. The anthropic approach also dissolves the problem, but is unfalsifiable. RS is falsifiable (see §**??**).

7 Predictions and Falsifiers

1. **No SUSY particles**: No superpartners at any energy. A confirmed discovery of a superpartner would indicate that the hierarchy is solved by SUSY rather than dissolved by the φ-ladder.

2. **No composite Higgs signals**: The Higgs coupling deviations from SM predictions should remain below the percent level. Large deviations would indicate compositeness.

3. **No extra dimensions**: No KK excitations or microscopic black holes at any collider energy.

4. **Standard Model desert**: Between the electroweak and Planck scales, no new particle species exist. The "great desert" is a structural prediction.

5. **Higgs mass stability**: The measured Higgs mass should remain consistent with the SM prediction from the φ-ladder, without additional corrections from new physics.

6. **Vacuum stability**: The electroweak vacuum is stable (not metastable), because the J-cost minimum is unique and global.

8 Conclusion

The hierarchy problem is dissolved in RS. The electroweak scale $v \approx 246$ GeV sits at rung $r_v \approx 27$ on the φ-ladder; the Planck scale at rung $r_P \approx 107$. The 80-rung gap is a structural feature of the ladder, not a fine-tuning. No new particles are required because there are no bare parameters to protect—masses are structural outputs of the cost minimization, not inputs subject to renormalization. The continuing null results at the LHC for SUSY, composite Higgs, and extra dimensions are predictions of the framework.

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